

YenCare Health Platform

Week 4 Task Assignments & QA Review Action Plan (Sprint Before Final Presentation)

Programme / Context	AmaliTech CSR Capstone Internship Programme — Product 4 (KNUST Health Services)
Active Sprint State	Gate 4 Milestone Fully Audited & Verified • Ready for Final Week (Gate 5)
Current Execution Date	Friday, 25 September 2026 (Gate 4 Audit & Sign-Off)
Sprint Goal	Enforce clinical policy (booking caps, double-booking guard), audit trails, SMS OTP, public rate limits, staff workstation polish, fail-fast env validator, and Playwright hermetic automation.
Automated Test Health	405 Passing Tests across 98 Suites (0 Failures, 100% Clean native Node Test Runner)
End-to-End Test Health	37 Passing, 11 Skipped, 0 Failing via Playwright in CI
Production Build Status	100% Clean Vite Production Bundles for both frontend and ui/prototype
Lead Author & Role	Blessing Edmund Kwame Dogbe (Backend & DevOps Lead / Product Coordinator)

1. Team Roster & Competency Allocation (6 Members)

We welcome Sterling Awuley as our Quality Assurance (QA) and Testing Lead, who will also assist with backend tasks. Engineering allocations for Week 4 align with core engineering strengths:

Team Member	Discipline	Key Delivered Artifacts	Week 4 Sprint Responsibilities
Blessing Edmund Kwame Dogbe	DevOps / Backend Lead	Render cloud staging, GitHub Actions CI automation, PR review/approval/merge governance, test scaling to 275 passing.	Sprint governance, startup env validator, RBAC permissions, architecture docs & runbooks.
Able Kafu Azanda	Backend Lead Architect	Virtual Queue Engine (PR #13), atomic daily tokens (A-01, B-04), Cancellation & Reschedule API (PR #14), slot release.	Comprehensive audit logging (actor type, old/new times), student booking cap policy (2-3 max).
Sterling Awuley (NEW MEMBER)	QA & Testing Lead	Authored PR #19 (queue sequence index) and PR #20 (visit-day guards in visitDayGuard.js). Clinical testing expert.	PR #20 manual clinical testing runbook (9 cases), investigate doctor-scoped booking visibility, past slot check.
Emmanuella Lodonu	Backend & Security	mNotify SMS sandbox wrapper, single-recipient safeguards, Call-Patient SMS trigger (PR #15), patient lookup API.	Cancellation/reschedule SMS OTP verification, lookup privacy (hide phones), public phone masking.
Raymond B. Afrani	Frontend Lead	Complete patient intake wizard (P01-P07), speakable YC reference, details/cancel UI (P11-P17), Queue Status (P18).	Staff login toast confirmation, prevent accidental back-button landing redirects, offline error UX banner, guard-aware disabled action button states, mobile responsiveness audit (320px+).
Harry “Ephraim” Nartey	Fullstack Foundations	Frontend HTTP API client integration, Fast-Track Walk-In Intake modal (Task S01/S02, PR #16), Multi-Room consultation (PR #18).	Fullstack assistance on staff workstation workflows, doctor-scoped consultation filter assistance, cross-device testing support.

2. Review Findings & Fix Status Summary (From Clinical Testing)

Category / Triage Area	Count	Current Resolution / Planned Assignment
Core Cards Fixed & Confirmed on Real DB	10	All 10 Week 4 engineering cards implemented, merged, and verified on local & Atlas MongoDB.
Clinical Policy & Visit-Day Guards	4	Student booking cap (max 2), 60m/15m arrival window, no past-slot booking, no-show slot release.

Category / Triage Area	Count	Current Resolution / Planned Assignment
Security & Anti-Abuse Hardening	4	Server-derived audit logs, SMS OTP verification, arrival brute-force rate limiter, phone masking.
Staff Workstation Polish	4	Login confirmation toast, back-button popstate trap, offline retry banner, accessible action tooltips.
DevOps & Automated Test Infrastructure	3	Fail-fast startup env validator (env.js), centralized RBAC matrix (staffAuth.js), connection pooling.
Audit Findings & Follow-up for Gate 5	3	Frontend ESLint cleanup (12 issues), generic recursive NoSQL operator middleware, Render staging host verification.

Week 4 Focus: Complete Hardening Ahead of Gate 5 Final Presentation!

- Week 4 is our final engineering sprint before the Presentation Week (Gate 5). All code must be hardened, tested by hand, and documented.
- Sterling Awuley will drive the manual clinical QA checklist across real MongoDB instances, verifying patient arrival windows, reception overrides, and slot release behaviors.
- Able Kafu Azanda and Emmanuella Lodonu will resolve the remaining backend security tasks (AuditLog actor tracking, student booking caps, and SMS OTP verification).
- Raymond Afrani will lead the staff and patient frontend screens, adding the login confirmation toast, back-button guard, offline roster error banner with retry, and disabled action buttons with tooltips.

3. Week 4 Assigned Tasks & Engineering Cards (With Assigned Prototype References)

[QA-01] PR #20 Clinical Verification & Manual Testing Runbook	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & QA)
Assignee & Support	Primary: Sterling Awuley (QA Lead) • Support: Blessing Edmund Kwame Dogbe
Assigned Prototype	S02 (Today Operations Dashboard), S03 (Appointments Roster), S04 (Student Record), S05 (Live Queue), P18 (Live Queue Status).
Rationale	PR #20 merged into main; all 9 clinical test scenarios validated in comprehensive testing runbook against MongoDB.
Scope & Goals	Execute the 9-item manual testing checklist from Section 10 of the review report against a real local/staging MongoDB instance. Validate that reception and patient check-in rules reject future bookings, arrival window (60m early / 15m late) triggers correctly, reception override allows late check-ins, and no-shows properly free time slots.
Dependencies	Blocked by: None • Blocks: Gate 5 clinical simulation. • Target: Wednesday, 23 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	• Execute and log results for all 9 scenarios in the Manual Testing Checklist (Verified in LIVE_TESTING_GUIDE.md). • Verify that 'now serving' room marker clears on QueueCounter when marked as no-show (Implemented in queueEngine.js:452). • Verify that TimeSlot is marked isBooked: false and appointmentId: null on real database after no-show (Implemented in queueEngine.js:445). • Document any UI/UX anomalies observed during reception check-in (Documented in Retrospective Section 5).

[BE-QA] Doctor-Scoped Booking Filter & Past Slot Prevention (Issues E & Problem 6)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & QA)
Assignee & Support	Primary: Sterling Awuley (Backend Assistant) • Support: Able Kafu Azanda
Assigned Prototype	S05 (Doctor Live Queue Board / Room Consultation), P04/P05 (Choose Date & Time Picker).
Rationale	Resolves reported issue where bookings showed under every doctor, and prevents bookings for slots that already started earlier today.
Scope & Goals	1. Investigate and fix doctor consultation filtering so appointments are strictly scoped to the clinician assigned to the room. 2. Integrate assertSlotNotInPast(slot) into the appointment booking endpoint (POST /api/appointments) so patients cannot book slots earlier today that have already passed.
Dependencies	Blocked by: None • Blocks: Doctor consultation workflow. • Target: Thursday, 24 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Doctor view displays only patients booked for their assigned clinicianId and roomId (Implemented in appointmentsRoutes.js:231 & queueRoutes.js:11). - POST /api/appointments rejects slots earlier today with 'That time slot has already started' (Implemented in assertRequestedSlotNotStarted). - Automated tests added asserting doctor-scoped filtering and past-slot booking rejection (7 HTTP tests + Playwright doctor-isolation.spec.js).

[SEC-BE] Public Arrival Endpoint Rate-Limiting & Anti-Brute-Force Guard (Issue D)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & Security)
Assignee & Support	Primary: Sterling Awuley + Blessing Dogbe • Support: Blessing Edmund Kwame Dogbe
Assigned Prototype	P09 (Find Appointment), P11 (Appointment Details), P18 (Queue Status).
Rationale	Prevents malicious brute-forcing of 4-digit reference codes (YC-XXXX) via the public /arrive endpoint.
Scope & Goals	Extend the public endpoint rate limiter (rateLimit.js) to cover POST /api/appointments/arrive and POST /api/appointments/:reference/arrive. Enforce maximum 5 check-in attempts per IP/reference per 15-minute window, returning HTTP 429 when exceeded.
Dependencies	Blocked by: None • Blocks: Public API security. • Target: Wednesday, 23 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Rate limiter active on POST /api/appointments/arrive and /:reference/arrive (Mounted in app.js:79-91). - Returns 429 Too Many Requests with retry-after header after 5 failed attempts (Enforced via arrivalReferenceKey). - Automated test added in rateLimit.test.js verifying arrival endpoint threshold (All 7 rate limit tests passing).

[BE-AUDIT] Comprehensive Cancellation & Reschedule Audit Logging (Issue A)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & Audit)
Assignee & Support	Primary: Able Kafu Azanda (Backend Lead) • Support: Blessing Edmund Kwame Dogbe
Assigned Prototype	P12/P13 (Cancel Flow), P14-P17 (Reschedule Flow), S04/S08 (Staff Change Appointment).
Rationale	Clinical compliance requires an immutable audit trail of who modified bookings and when.
Scope & Goals	Enhance AuditLog recordings in appointmentOps.js during cancel and reschedule actions. Explicitly record actorType ('PATIENT' vs 'STAFF'), actingUserId (staff ID or patient phone/studentIndex), oldDate, oldTime, newDate, newTime, and changeReason. Fix acting user resolution from auth tokens.
Dependencies	Blocked by: None • Blocks: Regulatory audit compliance. • Target: Thursday, 24 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- AuditLog schema captures actorType, actorId, oldDate, oldTime, newDate, newTime, and reason (Implemented in models/AuditLog.js). - Cancellation logs whether executed by patient via self-service or desk staff via portal (Server-derived in resolveActingUser). - Reschedule logs previous and new date/time coordinates (Captured in appointmentOps.js). - Automated tests verify audit record creation upon cancellation and reschedule (14 tests passing in appointmentAudit.test.js).

[BE-POLICY] Student Booking Cap Enforcement & Double-Booking Guard (Issue B)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & Business Logic)
Assignee & Support	Primary: Able Kafu Azanda (Backend Lead) • Support: Sterling Awuley
Assigned Prototype	P02 (Your Details), P06 (Review Booking), P08 (Slot Taken Clash).
Rationale	Prevents individual students from monopolizing clinic consultation slots.
Scope & Goals	Implement an active booking policy in bookAppointment.js: count existing active appointments (BOOKED, CHECKED_IN, WAITING) for the studentIndex on future dates. Cap active bookings at 2 per student. Reject additional bookings with a clear clinical policy message.
Dependencies	Blocked by: None • Blocks: Fair campus access. • Target: Thursday, 24 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Counts active non-completed, non-cancelled bookings per studentIndex (Implemented in countActiveBookingsForStudent). - Rejects booking request with HTTP 409 if student already has 2 active upcoming bookings (Enforced in assertStudentBookingCap). - Clear error message: 'You have reached the maximum of 2 active appointments. Please complete or cancel existing visits.' - Automated unit tests asserting cap rejection (18 tests in bookAppointmentHardening.test.js + Playwright booking-guards.spec.js).

[SEC-SMS] Cancellation/Reschedule SMS OTP Verification & Lookup Privacy (Issue C)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & Security)
Assignee & Support	Primary: Emmanuella Lodonu (Backend & Security) • Support: Blessing Edmund Kwame Dogbe
Assigned Prototype	P09 (Find Appointment / Lookup), P12 (Cancel Confirm), P16 (Reschedule Review).
Rationale	Prevents unauthorized cancellation of a student's appointment by someone who only knows their phone number.
Scope & Goals	1. Modify GET /api/appointments/lookup so it masks or omits the full patient phone number in public responses. 2. Implement an SMS OTP verification flow: before executing cancellation or reschedule, send a 4-digit code to the registered phone and require it in the PATCH payload.
Dependencies	Blocked by: None • Blocks: Patient security. • Target: Friday, 25 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Public lookup API returns masked phone (e.g. +233 24 **** 567) (Implemented in maskAppointmentForLookup). - POST /api/appointments/:id/request-cancel-otp sends 4-digit SMS OTP (Implemented in services/otpService.js). - PATCH /api/appointments/:id/cancel requires valid otpCode unless initiated by authenticated staff (Enforced via verified JWT req.staff). - Automated tests verify OTP generation, verification, and expiration (15 tests passing in appointmentOtp.test.js).

[SEC-DATA] Patient Data Privacy, Phone Masking & Security Audit	
Status & Milestone	COMPLETED (VERIFIED & SANITIZED) Week 4 (Hardening & Security)
Assignee & Support	Primary: Emmanuella Lodonu (Backend & Security) • Support: Sterling Awuley
Assigned Prototype	S10 (Public Display Board), P18 (Queue Status), S05 (Corridor TV).
Rationale	Ensures Ghana Data Protection Act (Act 843) compliance on waiting room display boards.
Scope & Goals	Audit all public screens, queue activity endpoints (GET /api/queue/activity), and display boards to ensure student names and phone numbers are never broadcast in plaintext. Sanitize all string inputs against NoSQL operators (\$gt, \$regex).
Dependencies	Blocked by: None • Blocks: Compliance sign-off. • Target: Friday, 25 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Public queue endpoints return masked identifiers (Token A-02, Index 2061****) (Implemented in queueEngine.js getClinicActivity, appointmentsRoutes.js maskAppointmentForLookup). - Input validation strictly types and sanitizes request parameters with defense-in-depth generic NoSQL operator middleware (Implemented in security.js nosqlSanitizer, mounted in app.js). - Security audit documentation aligned with codebase reality in docs/security/security-and-compliance.md.

[FS-UX] Staff Workstation Experience, Login Toast & Offline Banners (Section 7)	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Hardening & UX)
Assignee & Support	Primary: Raymond B. Afrani (Frontend Lead) • Support: Harry "Ephraim" Nartey (Fullstack)
Assigned Prototype	- S01: Staff Sign In (ui/wireframe/s01_sign_in_desktop_staff, ui/prototype/src/components/staff/S01SignIn.tsx) - S02: Today Operations Dashboard (ui/wireframe/s02_today_desktop_staff, ui/prototype/src/components/staff/S02Today.tsx) - S03: Appointments Roster (ui/wireframe/s03_appointments_desktop_staff, ui/prototype/src/components/staff/S03Appointments.tsx & frontend/src/components/staff/StaffRoster.jsx) - G01: Network Error & Offline Edge State (ui/wireframe/g01_network_error_mobile, frontend/src/components/ConnectivityBanner.jsx)
Rationale	Addresses reception usability issues identified in Section 7 of the clinical testing review. Assigned to Raymond B. Afrani for frontend implementation.
Scope & Goals	- Display short 'Signed in as Receptionist' (or Doctor/Admin) confirmation toast upon login (frontend/src/pages/StaffLogin.jsx). - Prevent accidental navigation back to public landing page via browser back button when logged in (frontend/src/components/RequireStaffAuth.jsx popstate lock). - Upgrade offline error display to 'Could not load appointment roster. [Retry]' with interactive retry button (frontend/src/hooks/useClinicRoster.js, StaffPortal.jsx, & DoctorWorkstation.jsx). - Align prototype simulator behavior in ui/prototype/src/components/staff/S01SignIn.tsx and S03Appointments.tsx.
Dependencies	Blocked by: None • Blocks: Staff UX polish. • Target: Thursday, 24 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Toast appears on successful staff authentication and auto-dismisses in 3 seconds ('Signed in as Receptionist') (StaffLogin.jsx). - Browser back button stays within authenticated staff portal view instead of jumping back to public landing page (RequireStaffAuth.jsx). - Offline/network failure displays 'Could not load appointment roster. [Retry]' button that re-triggers fetch (useClinicRoster.js). - Tested on Chrome, Safari, and mobile viewports.

[FE-A11Y] Guard-Aware Disabled Action States & Mobile Viewport Audit (Section 7 & WCAG)	
Status & Milestone	COMPLETED (REFINED FOR A11Y) Week 4 (Hardening & Frontend)
Assignee & Support	Primary: Raymond B. Afrani (Frontend Lead) • Support: Sterling Awuley
Assigned Prototype	S02 (Today), S03 (Appointments), S04 (Patient Detail) (Disabled Check In / No-Show buttons with tooltips), All Mobile Screens P01-P18 (320px viewport).
Rationale	Staff must clearly see why an action is blocked instead of encountering silent failures or confusing alerts.
Scope & Goals	1. Provide informative tooltip badges on 'Check In' and 'Mark No-Show' buttons explaining visitDayGuard rules while preserving native button interactivity for reception overrides. 2. Complete cross-device responsive testing ensuring all screens render cleanly down to 320px width with 44px+ touch targets.
Dependencies	Blocked by: None • Blocks: Reception & patient UX. • Target: Friday, 25 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Check-in button provides informative tooltip explaining visit day restriction (StaffRoster.jsx native title tooltips). - No-show button provides informative tooltip during grace period explaining restriction while retaining button clickability for overrides. - All patient and staff interfaces pass mobile responsive check (320px viewport). - Touch targets satisfy WCAG AA 44x44px standard.

[DEVOPS-CORE] Startup Environment Validator, Role Permissions & Documentation	
Status & Milestone	COMPLETED (VERIFIED) Week 4 (Official Gate 4 Milestone)
Assignee & Support	Primary: Blessing Edmund Kwame Dogbe (Lead) • Support: AmaliTech Evaluators
Assigned Prototype	System-wide architecture, Environment Config, RBAC matrix, documentation runbooks.
Rationale	Implements Section 8 suggestions: fail-fast startup config verification and centralized RBAC permissions.
Scope & Goals	1. Author a startup configuration validator (config/env.js) checking all required env vars (PORT, MONGO_URI, JWT_SECRET, MNOTIFY_KEY) with clear diagnostic errors. 2. Centralize role permissions into an RBAC matrix instead of ad-hoc role checks. 3. Update all docs/ guides in preparation for Gate 5 evaluation.
Dependencies	Blocked by: None • Blocks: Gate 4 submission & Gate 5 defence. • Target: Friday, 25 Sep 2026 (Verified 25 Sep)
Acceptance Criteria	- Server startup halts with formatted diagnostic error if critical environment settings are missing (backend/src/config/env.js). - Centralized hasPermission(user, 'APPOINTMENTS_CHECKIN') helper in staffAuth.js with matrix tests in rbac.test.js. - docs/ documentation updated: API catalog, database-ops.md, testing-guide.md, LIVE_TESTING_GUIDE.md. - Gate 4 Submission form and project retrospective drafted and audited.

4. Sterling Awuley's QA & Manual Testing Runbook (PR #20 9-Item Checklist)

#	Action / Scenario	Target Role & Test Payload	Expected Result
1	Check in future-dated booking	Reception desk checking booking dated tomorrow	Refused: 'This appointment is for [date]. Check-in opens on the day of the visit.'
2	Patient 'I've arrived' on future date	Patient self-check-in on booking dated next week	Refused with the same visit-day restriction message.
3	Patient arrives > 60m early	Patient tapping 'I've arrived' 75m before slot time	Refused: 'Check-in opens 60 minutes before your appointment time.'
4	Patient arrives within window	Patient tapping 'I've arrived' 30m before slot time	Allowed: Status updates to CHECKED_IN.
5	Patient arrives > 15m late	Patient self-check-in 20m after scheduled slot	Refused: 'You are past your appointment time. Please see reception.'

#	Action / Scenario	Target Role & Test Payload	Expected Result
6	Reception check-in for late patient	Reception desk checking in the patient from Case 5	Allowed: Reception override permits check-in on the appointment day.
7	Reschedule to slot earlier today	Rescheduling appointment to slot that already started	Refused: 'That time slot has already started. Please choose a later slot.'
8	Mark future booking as no-show	Reception marking future-dated booking as no-show	Refused: 'Cannot be marked as a no-show before the day of the visit.'
9	Mark awaiting patient as no-show	Marking WAITING or BOOKED (post-grace) patient	Allowed: Status transitions to NO_SHOW; time slot is freed; QueueCounter active pointer cleared.

5. Architectural Decisions & Consensus (From Review Section 9)

Review Question / Decision Point	Agreed Team Architecture Decision
1. Arrival window (60m early / 15m late)?	Confirmed. Matches KNUST Students' Clinic OPD guidelines. Centralized in visitDayGuard.js (EARLY_WINDOW_MINUTES = 60, LATE_GRACE_MINUTES = 15).
2. No-show grace period (slot + 15m)?	Confirmed. A patient who could still arrive cannot be declared a no-show while their self-arrival window remains open. Arrived patients can be marked on the day.
3. Booking limit per student?	Confirmed. Cap active future bookings at 2 per student to prevent slot hoarding across departments.
4. Cancellation/reschedule SMS OTP?	Confirmed for patient self-service. Authenticated desk staff can cancel/reschedule with an audit reason without OTP.
5. Staff account credential management?	Managed centrally by administrator. Seed accounts provided for staging demonstration.
6. Doctor-scoped visibility & notifications?	Doctors strictly view their assigned room/consultation queue. SMS notifications reserved for patients to manage SMS operational costs.

6. Forward Roadmap: Gate 5 Final Presentation (Fri 2 Oct 2026)

Presentation Deliverable	Primary Owner	Scope & Done Criteria
[DEMO] End-to-End Simulation Rehearsal	Sterling Awuley & All Team	Conduct multi-role rehearsals simulating full clinic morning: 5 bookings, 2 walk-ins, receptionist check-in, doctor triage, live SMS texts.
[DATA] Clinical Impact Wait-Time Modeling	Able & Emmanuella	Model simulated throughput improvements: quantify how virtual queuing reduces peak waiting room congestion by 65% at KNUST clinic.
[SLIDES] Executive Pitch Deck & Video	Raymond & Harry	Produce final presentation slide deck, architecture diagrams, and 3-minute high-definition product walkthrough video.
[LEAD] Final Capstone Defence & Handover	Blessing Dogbe	Lead final presentation to AmaliTech CSR evaluators and KNUST clinical stakeholders. Deliver final repository bundle.